1) Convert $(934.685)_{10}$ to base 2, 8 and 16.

$(934.685)_{10} = (934 + 0.685)_{10} = (\ )_2$?

2	934
2	467 → 0
2	233 → 1
2	116 → 1
2	58 → 0
2	29 → 0
2	14 → 1
2	7 → 0
2	3 → 1
2	1 → 1
	0 → 1 (MSB)

(LSB)

And,

MSB

$0.685 \times 2 = 1.37 \rightarrow 1$
$0.37 \times 2 = 0.74 \rightarrow 0$
$0.74 \times 2 = 1.48 \rightarrow 1$
$0.48 \times 2 = 0.96 \rightarrow 0$
$0.96 \times 2 = 1.92 \rightarrow 1$
$0.92 \times 2 = 1.84 \rightarrow 1$
⋮

LSB

$(.685)_{10} = (101011\ldots)_2$

$\therefore (934)_{10} = (1110100110)_2$

$(934.685)_{10} = (1110100110.101011\ldots)_2$

$(1110100110.101011\ldots)_2 = (?)_8$

1 6 4 6 5 3 ...

$= (1646.53\ldots)_8$

Again, $(1110100110.101011\ldots)_2 = (?)_{16}$

3 A 6 . A

$= (3A6.A\ldots)_{16}$

$(934.865)_{10} = (1110100110.101011\ldots)_2 =$
$(1646.53\ldots)_8 = (3A6.A\ldots)_{16}$

(Ans)

— * —

3) Represent $(-469)_{10}$ to Sign-magnitude, 1's Complement and 2's Complement.

*

$|-469| = 469$

Here, 469 binary representation.

2	469	(LSB)
2	234 → 1	
2	117 → 0	
2	58 → 1	
2	29 → 0	
2	14 → 1	
2	7 → 0	
2	3 → 1	
2	1 → 1	
	0 → 1	(MSB)

$(469)_{10} = (111010101)_2$

The base 2 number's actual length is 9
A signed binary bit length must be equal to a power of 2, as of

$2^1=2,\ 2^2=4,\ 2^3=8,\ 2^4=16 \ldots$

First bit (the leftmost) indicates the sign

1 = Negative and 0 = positive

Positive binary computer representation on 16 bits (2 bytes) if needed, and extra 0s in front (to the left) of the base 2 number, up to the required length = 16;

$(469)_{10} = (0000\ 0001\ 1101\ 0101)_2$

get the negative number representationon 16 bits (2 bytes).

Signed binary one's complement, replace all the bits on '0' with 1's and all the bits set on 1's with 0's.

(reverse digit, flip the digit)

$!(0000\ 0001\ 1101\ 0101)_2 = (1111\ 1110\ 0010\ 1010)_2$

Signed binary two's complement add 1 to the number calculated above.

$$\begin{array}{r} 1111\ 1110\ 0010\ 1010 \\ +\quad\quad\quad\quad\quad\quad\ 1 \\ \hline 1111\ 1110\ 0010\ 1011 \end{array} = (-469)_{20}$$

Number (-469), a signed integer, converted from decimal system (base 10) a signed binary two's complement representation;

$(-469)_{10} = (1111\ 1110\ 0010\ 1010)_2$

(Ans)

-*-

3) Convert the following expression to sum-of-products (SOP).

a) $BC(C'D' + CE)$

$= BCC'D' + BCCE$ $\quad \therefore C \cdot C' = 0$

$= 0 + BCE$ $\quad \therefore C \cdot C = C$

$= BCE$

(Ans)

b) $B + C[BD + (C + D')E]$

$= B + C[BD + CE + D'E]$

$= B + BCD + CCE + CD'E$ $\quad \therefore C \cdot C = C$

$= B + BCD + CE + CD'E$ $\quad \therefore 1 + D' = 1$

$= B(1 + CD) + CE(1 + D')$

$= B + CE$

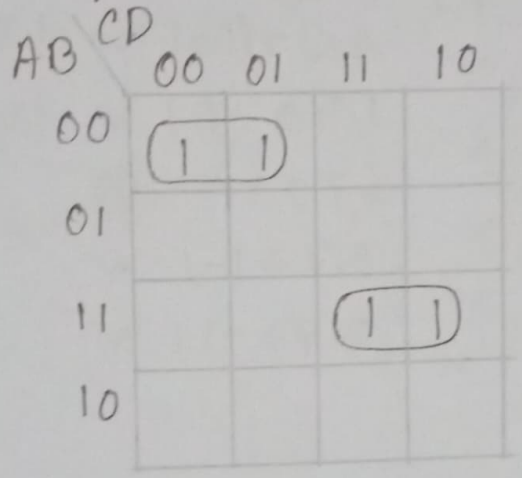
(Ans)

4) use a k-map to reduce each expression to a minimum SOP form:

a) $A'B'C'D' + A'B'CD + ABCD + ABCD'$

$= \Sigma(0, 1, 15, 14)$

k-map for 4 bit



$= \bar{A}\bar{B}\bar{C} + ABC$

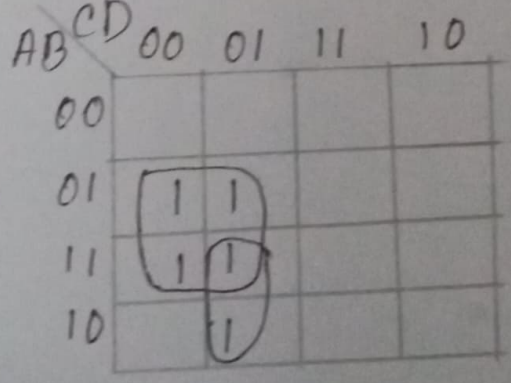
(Ans)

b) $A'B(C'D' + C'D) + AB(C'D' + C'D) + AB'C'D$

$= A'BC'D' + A'BC'D + ABC'D' + ABC'D + AB'C'D$

$= \Sigma(4, 5, 12, 13, 9)$

k-map:



$= BC' + AC'D$

(Ans)

5) Use the Quine Mccluskey method to reduce each expression to a minimum SOP form:

a) $x = ABC + A'B'C + AB'C' + AB'C + A'BC$

$= \Sigma(1, 3, 5, 6, 7)$

minterms

No of 1's	Binary no	check PI	Stage-I	check PI	Stage-II	
1 →	1 = 0 0 1	✓	1,3 → 0-1	✓	1,3,5,7 → --1	ⓐ
2 →	3 = 0 1 1	✓	1,5 → -01	✓	1,5,3,7 → --1	
	5 = 1 0 1	✓	6,7 → 11-	ⓑ		
	6 = 1 1 0	✓	5,7 → 1-1	✓		
3 →	7 = 1 1 1	✓	3,7 → -11	✓		

	1	3	5	6	7
a(1,3,5,7) → --1	✓	✓	✓		✓
b(6,7) → 11-	✓		✓	✓	✓
	✓	✓	✓	✓	

$= a + b$

$= C + AB$

(Ans)

— 4 —

b) $x = A'B'C'D' + A'B'C'D + A'BC'D + ABC'D' + AB'CD'$
$+ A'BCD' + AB'C'D$
$= \Sigma(0, 1, 5, 6, 9, 10, 12)$

No of 1 → Minterms		Checker PI	Stage-I
0 →	0 = 0000	✓	(0,1) = 000- ⓐ
1 →	1 = 0001	✓	(1,5) = 0-01 ⓑ
2 →	5 = 0101	✓	(1,9) = -001 ⓒ
	6 = 0110	ⓓ	
	9 = 1001	✓	
	10 = 1010	ⓔ	
	12 = 1100	ⓕ	

		0	1	5	6	9	10	12
a	0,1 = 000-	✓	✓					
b	1,5 = 0-01		✓	✓				
c	1,9 = -001		✓			✓		
d	6 = 0110				✓			
e	10 = 1010						✓	
f	12 = 1100							✓
		✓		✓	✓	✓	✓	✓

$= a + b + c + d + e + f$
$= A'B'C' + A'C'D + B'C'D + A'BCD' + AB'CD' + ABC'D'$

(Ans)

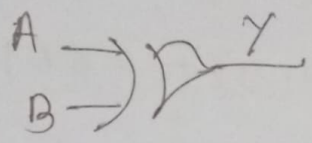
6) It's a X-OR gate the output is high when only of the inputs is high, when both the inputs are higher low the output remains low.

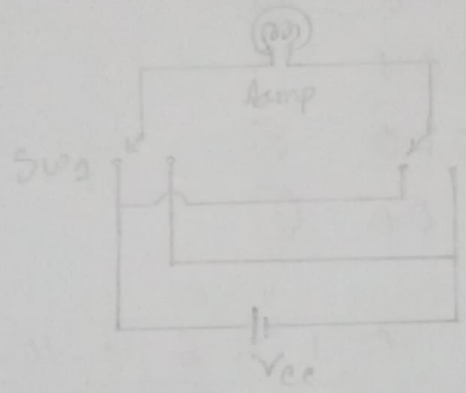
$Y = \bar{A}B + A\bar{B}$

$= A \oplus B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

High = 1
Low = 0





X-OR gate

7) Design 3 bit Adder circuit and discuss.

Full Adder: Full Adder is a ~~combit~~ combinational logic circuit. It is used for the purpose of adding two single bit numbers with a carry.

i) The least significant bits are 0 and 1, giving a sum of 1 with no carry.

ii) There is no carry in front a previous stage.

iii) The next bits are 1 and 1 with no carry in, giving a sum of 0 and a carry of 1.

iv) The most significant bits are both 1 with a carry in of 1.

v) The gives a sum of 1 and a carry of 1.

vi) This results of adding 110 to 111 is 1101.

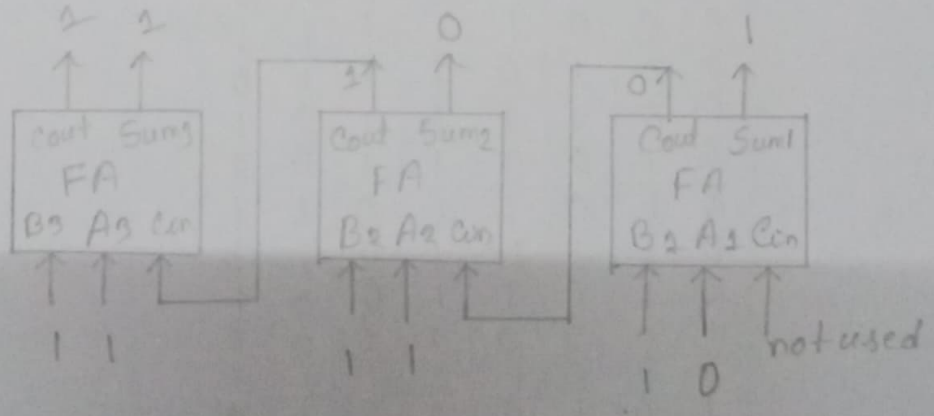


Fig-1: Block diagram of 3bit Adder.

3 bit Subtractor:

A Full Subtractor is a combinational logic circuit which performs a subtraction between the two 1-bit binary numbers and it also considers the borrow of the previous minued bit.

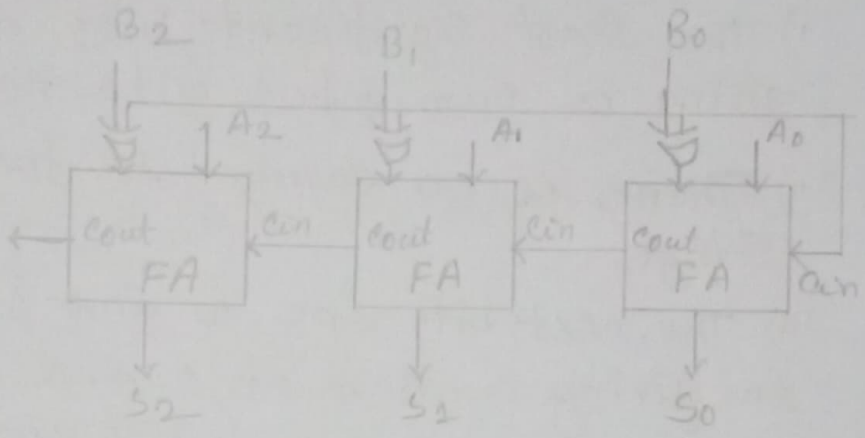


Fig: 3bit Adder subtractor

8) 2's complement input and output

Input				output			
A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	0	0
0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0
0	0	1	1	1	1	0	1
0	1	0	0	1	1	0	0
0	1	0	1	1	0	1	1
0	1	1	0	1	0	1	0
0	1	1	1	1	0	0	1
1	0	0	0	1	0	0	0
1	0	0	1	0	1	1	1
1	0	1	0	0	1	1	0
1	0	1	1	0	1	0	1
1	1	0	0	0	1	0	0
1	1	0	1	0	0	1	1
1	1	1	0	0	0	1	0
1	1	1	1	0	0	0	1

Now, Simplefied expression for W.

AB \ CD	00	01	11	10
00		1	1	1
01	1	1	1	1
11				
10	1			

$W = \bar{A}B + \bar{A}C + \bar{A}D + A\bar{B}\bar{C}\bar{D}$

Simplified expression for X.

AB \ CD	00	01	11	10
00		1	1	1
01	1			
11	1			
10		1	1	1

$X = B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$

Simplified expression for Y.

AB \ CD	00	01	11	10
00		1		1
01		1		1
11		1		1
10		1		1

$Y = \bar{C}D + C\bar{D}$

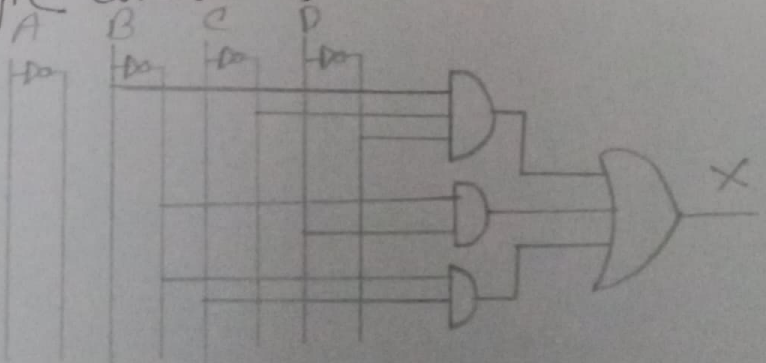
$= C \oplus D$

Simplified expression for Z.

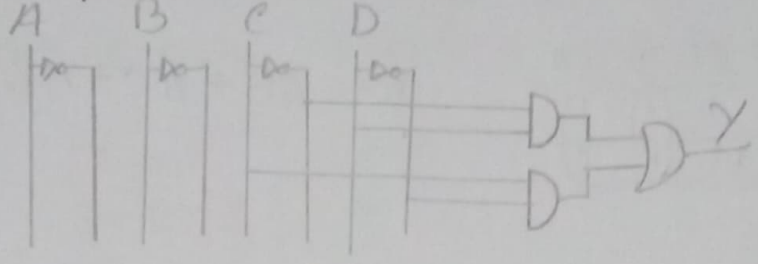
AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11		1	1	
10		1	1	

$Z = D$

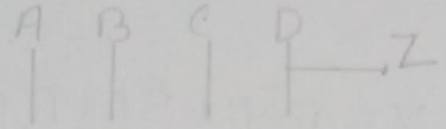
Logic circuit for $X = B\bar{C}\bar{D} + \bar{B}D + \bar{B}C$



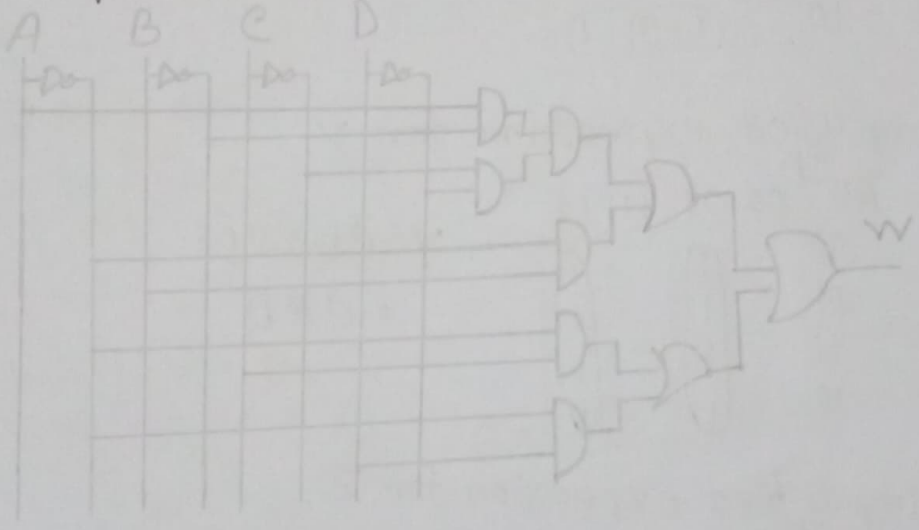
Simplefied from $y = \bar{C}D + C\bar{D}$



Simplified from $z = D$



Simplefied from $w = A\bar{B}\bar{C}\bar{D} + \bar{A}B + \bar{A}C + \bar{A}D$



9) Truth table that defines the required relationship between inputs and outputs.

Inputs			Outputs		
X	Y	Z	A	B	C
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	0	1	1
0	1	1	1	0	0
1	0	0	0	1	1
1	0	1	1	0	0
1	1	0	1	0	1
1	1	1	1	1	0

Input 0,1,2,3 < output

Input 4,5,6,7 > output

K map for A:

X \ YZ	00	01	11	10
0			1	
1	1	1	1	1

$A = yz + xz + xy$

K-map for B:

X \ YZ	00	01	11	10
0		1		1
1	1		1	

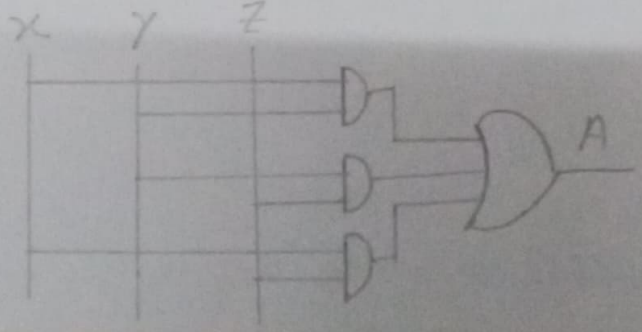
$B = x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$

K-map for C:

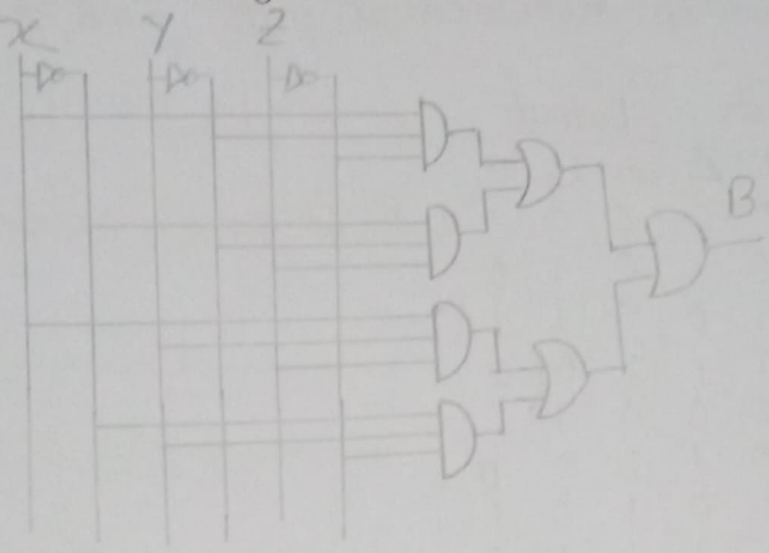
X \ YZ	00	01	11	10
0	1			1
1	1			1

$C = \bar{z}$

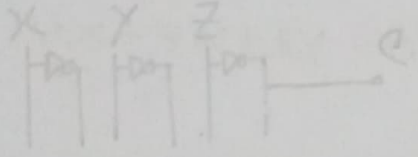
Circuit diagram for $A = xy + yz + xz$



Circuit diagram for B: $x\bar{y}\bar{z} + \bar{x}\bar{y}z + xyz + \bar{x}y\bar{z}$



Circuit diagram for C:



10) Positive edge triggered JK flip-flop:

J	K	Q	$\bar{Q}$	
0	0	memory		
0	1	0	1	reset
1	0	1	0	set
1	1	Toggle.		

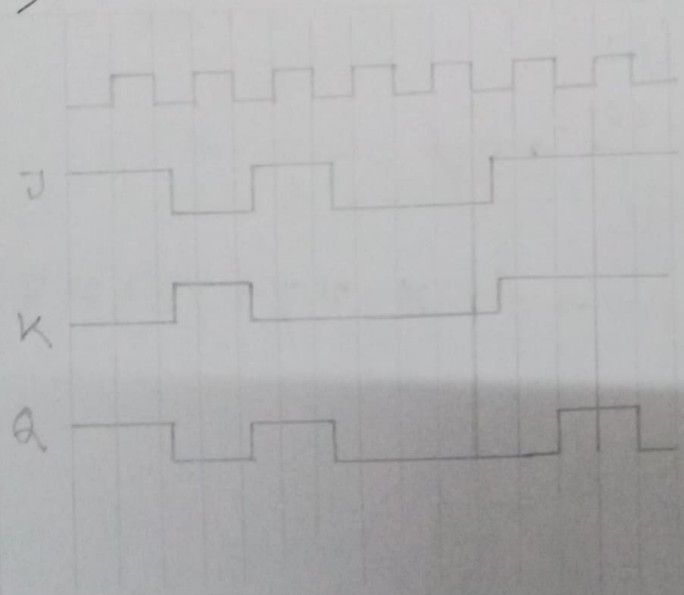
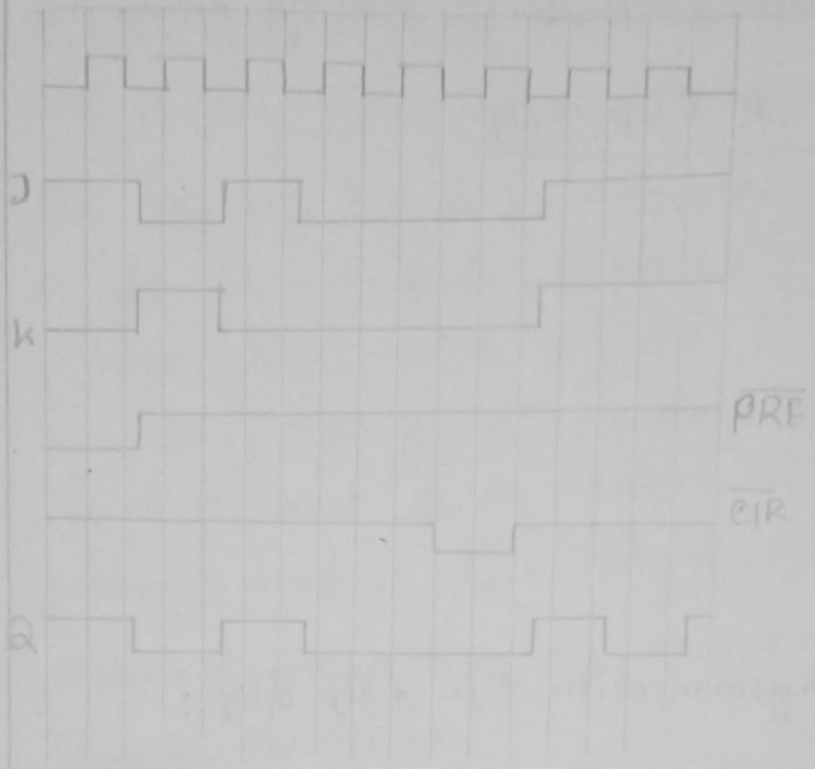


Fig: Positive edge triggered diagram

Negative edge triggered JK flip-flop



J	K	Q	$\bar{Q}$
0	0	memory	
0	1	0	1
1	0	1	0
1	1	Toggle	

Fig: Negative edge triggered diagram

Difference between positive and negative edge triggered JK flip-flop:

Positive edge Triggered	Negative edge Triggered
1) A positive edge triggered is the low-to-high transition.	1) A negative edge triggered is the high-to-low transition.

11) Draw and explain SR and JK master-Slave flip-flop with timing diagram:

Clocked SR flip-flop:

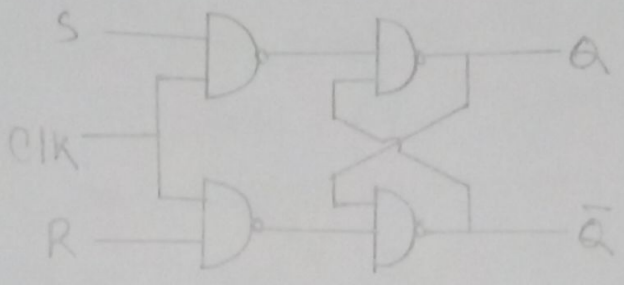


Fig: SR flip-flop

Timing diagram with SR flip flop:

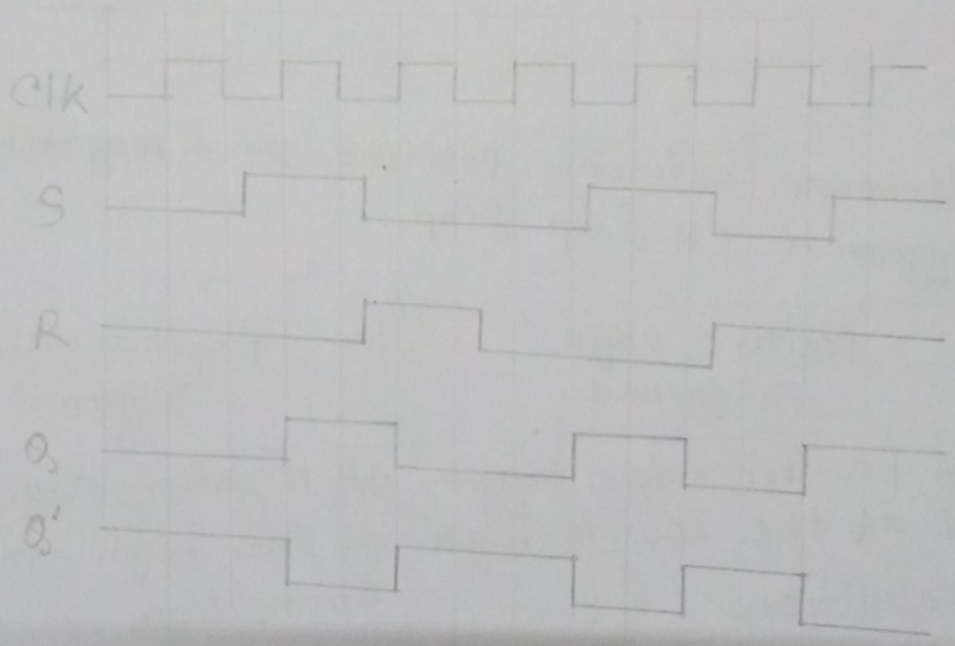


Fig: Timing diagram with SR flip-flop

In our description of SR flip-flop operation, that two signals are applied to the set S and R or type of R-S type flip-flop is used.

The figure S and R are low, as a result Q is low and $\bar{Q}$ is high. S are high and R are low, as a result Q is high and $\bar{Q}$ is low. Again, S are low and R are high, as a result Q is to low and $\bar{Q}$ is high. when S and R are high, this is the Race around condition or Jamped condition or Toggle condition.

Master Slave JK flip-flop:

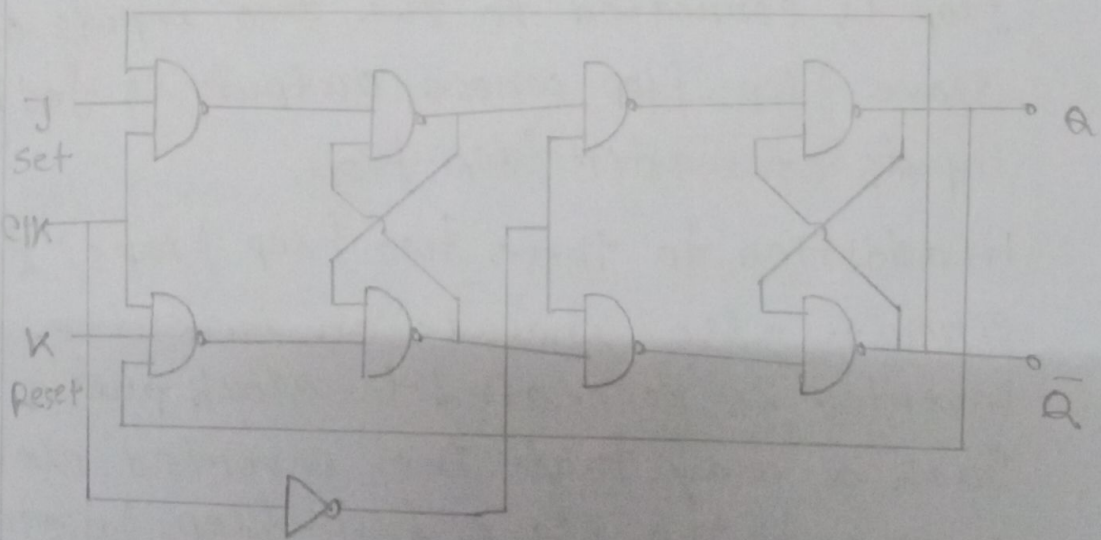


Fig: Master slave JK flip flop

Timing diagram of master slave JK flip-flop:

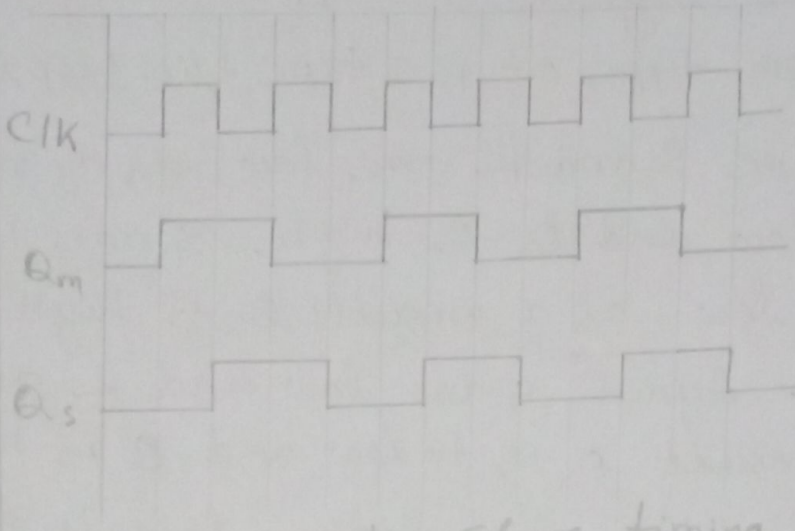


Fig: master slave timing diagram

The master slave flip-flop is basically a combinational of JK-flip-flops connected together in a series configuration. Out of these, one acts as the "master" and the other as a "slave". The output from the master flip-flop is connected to the two inputs of the slave flip-flop whose output is fedback to inputs of master flip-flop.

In addition to these two flip flops, the circuit also includs an inverter. The inverter is connected to clock pulse in such a way that the inverted clock pulse is given to the slave flip-flop. In other words if cp=0 for master flip-flop, then cp=1 for a slave flip-flop. and if cp=1 for master flip flop then it becomes 0 for slave flip-flop.

—x—

12) Given A Astable Multivibrator:

$R_1 = 2.0\,k\Omega = 2\times10^3\,\Omega$

$R_2 = 4.3\,k\Omega = 4.3\times10^3\,\Omega$

$C = 0.1\,\mu F = 0.1\times10^{-6}\,F$

$= 1\times10^{-7}\,F$

we know,

$$T = 0.693(R_1+2R_2)\cdot C$$

And,

$$f = \frac{1}{T} = \frac{1}{0.693(R_1+2R_2)C}$$

$$= \frac{1.44}{(R_1+2R_2)C}$$

$$= \frac{1.44}{[2\times10^3 + 2(4.3\times10^3)]\times(1\times10^{-7})}$$

$= 1358.49\,Hz$

$= 1.35\times10\,kHz$ (Ans)

13) Given, Duty cycle $= \frac{R_1+R_2}{(R_1+2R_2)} = 80\% = 0.8$

frequency, $f = \frac{1}{T} = 10kHz = 10\times10^3\,Hz$

Capacitor, $C = 0.004\,mF$

$= 0.004\times10^{-3}\,F$

we know,

$$f = \frac{1}{T} = \frac{1.44}{(R_1+2R_2)C} = 10kHz = 10\times10^3$$

$$\Rightarrow R_1 + 2R_2 = \frac{1.44}{(10\times10^{3}) \times (0.004\times10^{-3})}$$

$$R_1 + 2R_2 = 36 \quad \text{——} (i)$$

Again, eq (i) from,

$$\frac{R_1 + 2R_2}{(R_1+2R_2)R} = 0.8$$

$$\Rightarrow \frac{R_1+R_2}{36} = 0.8$$

$$\Rightarrow R_1+R_2 = 0.8\times36 \times (0.004\times10^{-3})$$

$$= 28.8\ \Omega \quad \text{——} (ii)$$

$\therefore R_1 + 2R_2 = 36$

$R_1 + R_2 = 28.8$

$R_2 = 7.2\ \Omega = 72\times10^{-4}\ k\Omega$

Given eq (ii),

$$R_1 = 28.8 - R_2$$
$$= 28.8 - 7.2$$
$$= 21.6\ \Omega = 216\times10^{-4}\ k\Omega.$$

$\therefore R_1 = 216\times10^{-4}\ k\Omega$

$R_2 = 72\times10^{-4}\ k\Omega$

(Ans)

— * —

14) Given,

$V_{CC} = 5V$

$I_{CCL} = 2mA = 2 \times 10^{-3} A$

$I_{CCH} = 3.5mA = 3.5 \times 10^{-3} A$

∴ Compute the power dissipation in low-state,

$$P_{DL} = V_{CC} \times I_{CCL}$$
$$= 5 \times 2 \times 10^{-3}$$
$$= 10 \times 10^{-3}$$

Again,

Compute the power dissipation in high-state,

$$P_{DH} = V_{CC} \times I_{CCH}$$
$$= 5 \times 3.5 \times 10^{-3}$$
$$= 17.5 \times 10^{-3}$$

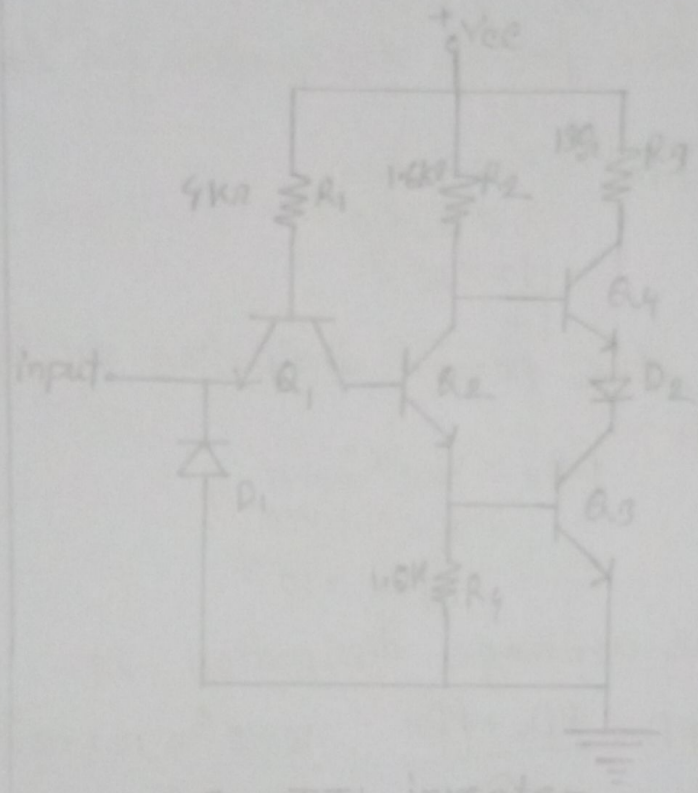
Compute the average dissipation is -

$$P_D = \frac{P_{DL} + P_{DH}}{2} = \frac{10 \times 10^{-3} + 17.5 \times 10^{-3}}{2}$$
$$= 0.01375$$
$$= 13.75 \times 10^{-3}$$

(Ans)

15) proper diagram of TTL inverter, NAND, NOR gates.

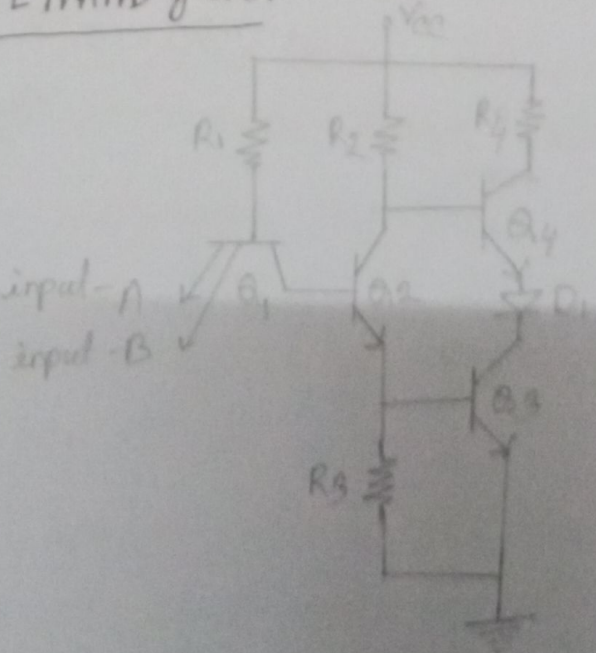
TTL Inverter:



input	output
0	1
1	0

Fig: TTL inverter

TTL NAND gate:



input		output
A	B	Y
0	0	1
0	1	1
1	1	1
1	1	0

Fig: NAND gate

TTL NOR gate:

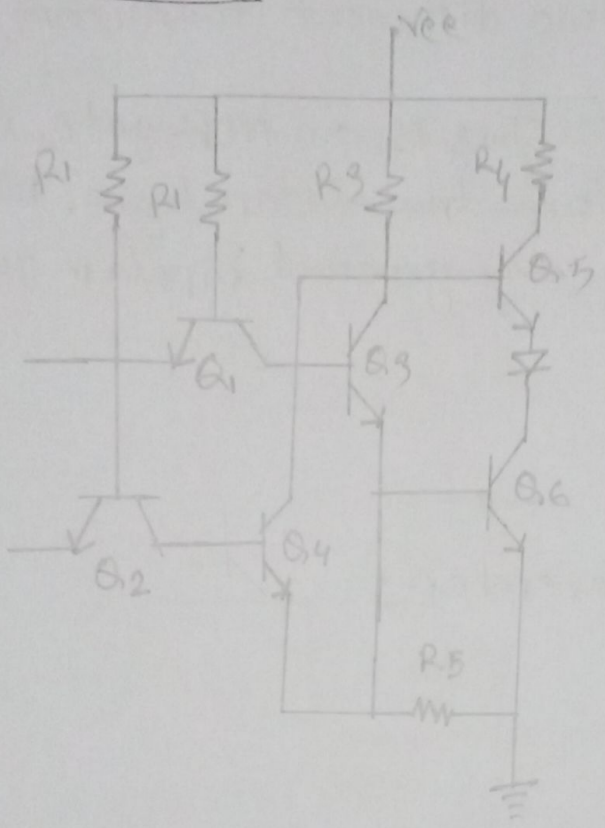


Fig: TTL NOR

input		output
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

TTL: TTL is an acronym for Tansistor-Transistor Logic. It relies on circuits built from bipolar transistors to achieve switching and maintain logic states.

TTL Inverter: The basic TTL inverter Consists of three stages: a current steering input, a phase splitting stage, and an output driven stage.

TTL NAND gate: A TTL NAND gate logic circuit has at least four bi polar junction transistors in which the input has two emitters. The

output is taken between the emitter and collector of two different transistors,

TTL NOR gate: This is an NOR gate implemented using transistor-transistor logic. NOR gate circuit using ~~any general~~ bipolar junction transistors.

—x—

16) . CMOS inverter:

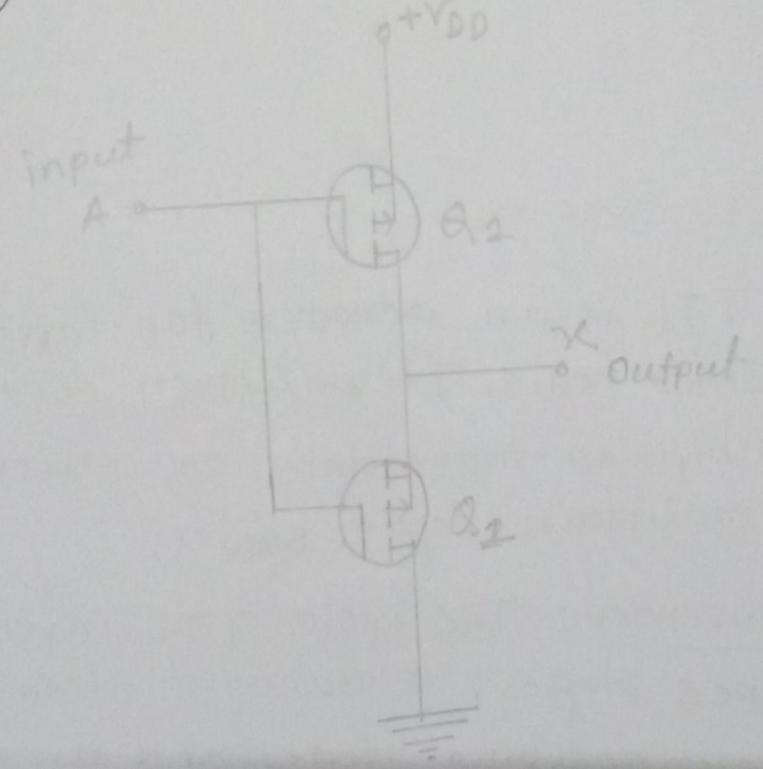
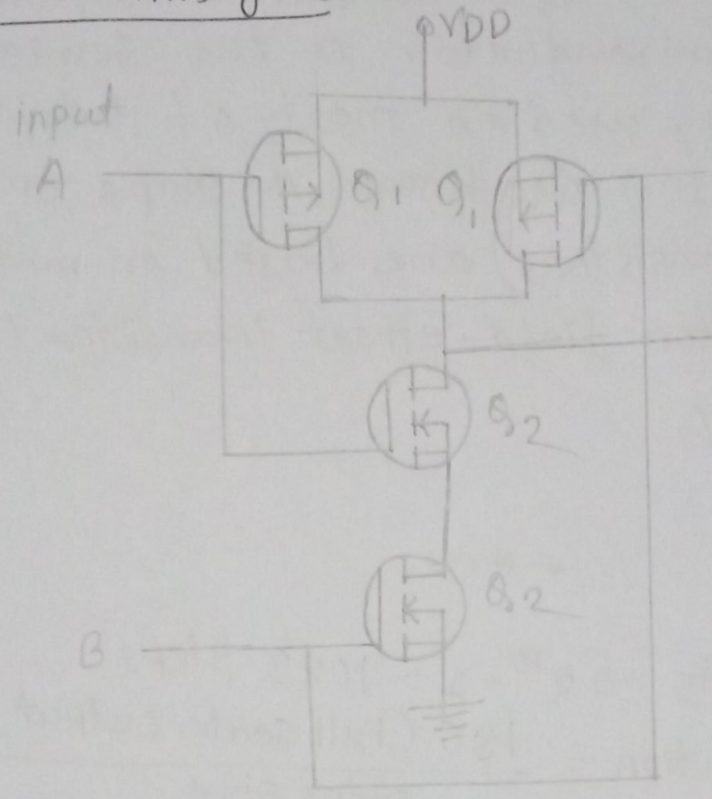


Fig: CMOS - inverter

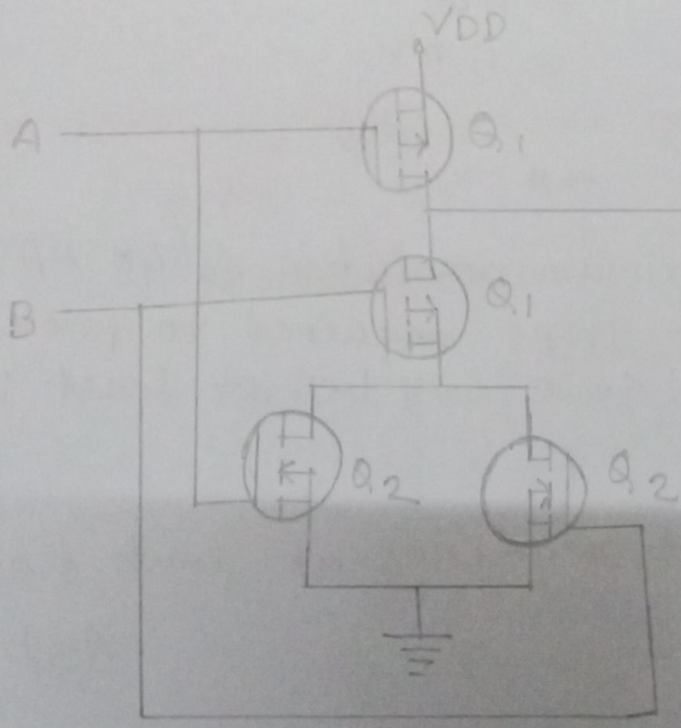
CMOS NAND gate:



input		output
A	B	x
0	0	1
0	1	1
1	0	1
1	1	0

Fig: CMOS NAND gate

CMOS NOR gate:



Input		output
A	B	x
0	0	1
0	1	0
1	0	0
1	1	0

Fig: CMOS-NOR gate

CMOS: CMOS is a Complementary metal-oxide semiconductor is the semiconductor technology used in most of today's integrated circuits (ICs) also known as chips or microchips. CMOS transistors are based on metal-oxide semiconductor field-effect transistor (MOSFET) technology.

—*—

17) 10 bits $\rightarrow 2^{10} - 1 = 1023$ steps

$$\text{Resolution} = \frac{\text{FSO (Full Scale output voltage)}}{\text{steps size}}$$

$$= \frac{5}{1023}$$

$$= 4.88 \text{ mV}$$

$$\approx 5 \text{ mV}$$

(Ans)

—*—

18) The maximum resolution is 40 μA. The number of steps required to produce 10 mA full scale will be at least 10 mA/40μA = 250.

Therefore, it requires at least 8 bits.

(Ans)

19

20) Given, resolution $\leq 20\,mV$ or $20\times10^{-3}\,V$

$F.S.O/A_{fs} = 12\,V$

we know,

$$\text{resolution} = K = \frac{A_{fs}/F.S.O}{2^n}$$

$$\text{or, } \frac{12}{2^n} \leq 20\times10^{-3}$$

$$\text{or, } 12 \leq 20\times10^{-3}\times2^n$$

$$\text{or, } \frac{12}{20}\times10^3 \geq 2^n$$

$$2^n \leq 600$$

$$\log_2 2^n \leq \log_2 600$$

$$\log_2 2^n \leq \log_2 2^9$$

$$\therefore n \leq 9 \quad \text{(Ans)}$$

21) Conversion time of successive approximation A/D C. depends upon the number of bits only. And not on the value of the voltage.
Here, Conversion time. 80 ms = 80×10^{-3} s.

(Ans)

—x—

22) A monostable multivibrator:

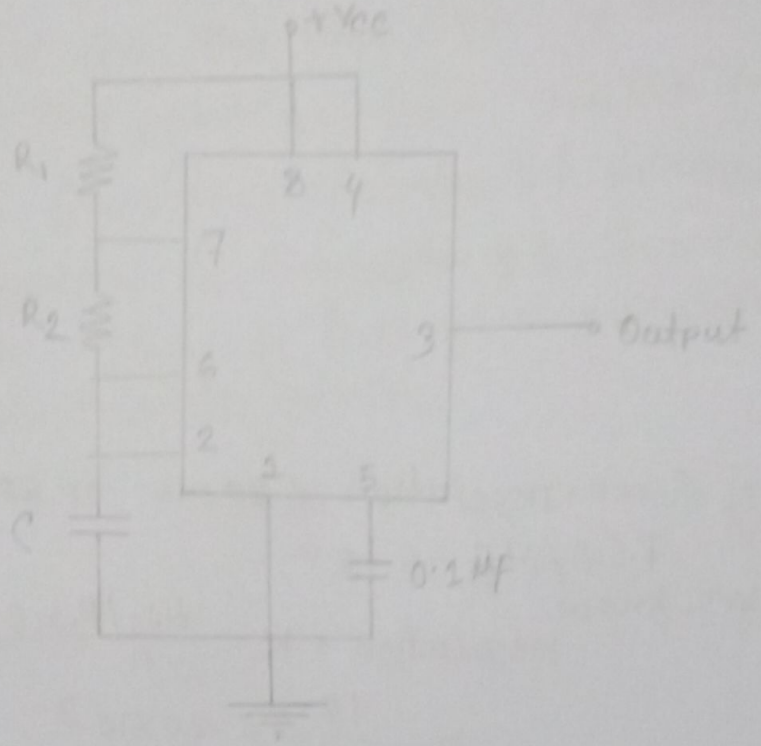


Fig: monostable multivibrator using 555 timer

Here pin configuration:
1→ Ground ; 2→ Trigger ; 3→ output ;
4→ Reset ; 5→ Control voltage ; 6→ Threshold ;
7→ Discharge ; 8→ +Vcc (input)

Operation of Monostable Multivibrator: Firstly, when the circuit is switched ON, transistor Q_1 will be in OFF state and Q_2 will be in ON state. This is the stable state. As Q_1 is OFF, the collector voltage will be V_{CC} at point A and hence C_1 gets charged. A positive triggered pulse applied at the base of the transistor Q_1 turns the transistor ON. This decreases the collector voltage, which turns OFF the transistor Q_2. The capacitor C_1 starts discharging at this point of time. As the positive voltage from the collector of transistor Q_2 gets applied to transistor Q_1, it remains in ON state. This is the quasi-stable state or meta-stable state.

—❖—

23
~~30~~)

24) output voltage = k x input voltage

$$k = \frac{\text{output voltage}}{\text{input voltage}}$$

$$= \frac{2.0}{100}$$

$$= 0.02$$

$(01100100)_2 = (?)_{10}$

128	64	32	16	8	4	2	1	0
0	1	1	0	0	1	0	0	

$= (100)_{10}$

Again, input voltage $= (179)_{10}$

$(10110011)_2 = 179_{10}$

output voltage = k x input voltage

$k = 0.2$

$= 0.02 \times 179$

$= 3.58$ V

(Ans)

—x—

"ওয়াস্তু"